

Aidan P. LaBella

Email : alabella@brown.edu

Citizenship: United States of America

<https://github.com/aidanlabella/>

<https://cs.brown.edu/people/grad/alabella/>

Mobile (Preferred) : +1 (518) 779-9285

INTERESTS

Data Science and Engineering • Machine Learning • Computational Science • Time Series Analysis

EDUCATION

- **Ph.D., Computer Science, Brown University** Providence, RI
2023 - Present
In Progress
 - **Advisor:** Stephen Bach
- **Sc.M., Computer Science, Brown University** Providence, RI
2023 - 2025
Conferred May 2025
 - **Examination Committee:** Stephen Bach, Karianne Bergen, Jonathan Pober
- **B.Sc. with Honors, Computer Science, Rochester Institute of Technology (RIT)** Rochester, NY
2018 - 2023
Conferred May 2023
 - **Advisor:** Travis Desell
 - **Minor and Immersion:** Criminal Justice

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, C#, Bash, ECMAScript, GoLang, R, MATLAB, Common Lisp, Icon, Prolog.
- **Tools:** Git, Maven, Vim, Visual Studio, jQuery, React, Bootstrap, Docker, Atlassian JIRA, Jenkins.
- **DBMS:** MySQL, MariaDB, MongoDB, PostgreSQL, SQLite.
- **Cloud:** AWS, Kubernetes.

EXPERIENCE

- **Undergraduate Research Assistant** RIT, Rochester, NY
Distributed Data Science Systems Lab May 2020 - May 2023
 - Worked with faculty and graduate students in research projects focused on Aviation Safety and Machine Learning.
 - Co-authored journal and conference articles covering the research conducted.
 - Trained new students coming into the lab that would be working on our various projects.
- **Software Engineer Co-Op** EagleView Technologies, Rochester, NY
Full member of an Agile Release Train (ART) and participant in the Agile SDLC Jan. 2022 - Apr. 2022
 - Full-stack software developer as part of an Agile team responsible for delivering EagleView's imagery to customers via web-based applications.
 - Performed incremental front-end changes in JavaScript for EagleView's main software platform, ConnectLink.
 - Utilized GoLang on backend services to ensure server reliability.
- **Peer Mentor** RIT, Rochester, NY
Tutored students in the introductory Computer Science sequence at RIT Jan. 2020 - Apr. 2020
 - Advised and mentored students with programming assignments, exam preparation, problem solving and technical issues.
 - Worked in both an in-person and online format to provide assistance to students.
 - Provided assistance for Java, C, C++ And Python languages.

PROJECTS

CURRENT

- **“SWIFT-SAT: RFI Detection Across Six Orders of Magnitude in Intensity: A Unifying Framework with Weakly Supervised Machine Learning”**: *Funded by the National Science Foundation (Award #2228989)*
 - Develop machine learning applications for the detection and removal of radio frequency interference (RFI) from 21cm cosmology data.
 - Introduce weak supervision to harvest RFI detection rules that produce meaningful training datasets.
 - Interdisciplinary work with members of Brown’s Computer Science and Physics departments.

PAST

- **The National General Aviation Flight Information Database (NGAFID)**: *Funded by the Federal Aviation Administration*
 - Implemented calculations for Loss of Control and Stall Indices. Both of these indices are utilized by pilots to identify potentially unsafe flight conditions.
 - Performed numerous UI/UX enhancements, bugfixes and added features at the request of the project stakeholders.
 - Added features that enabled users to simulate previous flights in the X-Plane Flight Simulator.
- **Evolutionary eXploration of Augmenting Memory Models (EXAMM)**:
 - Implemented a backpropagation-free neural architecture evaluation method known as Random Error Sampling-based Neuroevolution (RESN).
 - Trained recurrent neural networks in predicting various time series columns from the NGAFID project data.

PUBLICATIONS

- [1]: Aidan LaBella, Joshua Karns, Farhad Akhbardeh, Andrew Walton, Zechariah Morgan, Brandon Wild, Mark Dusenbury and Travis Desell. **Optimized Flight Safety Event Detection in the National General Aviation Flight Information Database** *In The 37th ACM/SIGAPP Symposium on Applied Computing (SAC '22)*. Online. April 25-29, 2022.
- [2]: Hong Yang, Aidan LaBella and Travis Desell. **Predictive Maintenance for General Aviation Using Convolutional Transformers** *The Thirty-Fourth Annual Conference on Innovative Applications of Artificial Intelligence (IAAI-22)*. Vancouver, British Columbia, Canada. February 24 - 26, 2022.
- [3]: Aidan LaBella, Charles Duong, Pak Iong Long, Nathan DePiero, Aditya Iyer, Elise Carman, Randall Balestrieri and Travis Desell. **GATS: A Time-Series Dataset for Addressing General Aviation Flight Safety** *1st ICML Workshop on Foundation Models for Structured Data*

PUBLIC TALKS

- Aidan LaBella. **Optimized Flight Safety Event Detection in the National General Aviation Flight Information Database**. *The 37th ACM/SIGAPP Symposium on Applied Computing (SAC '22)*. Online. April 25-29, 2022.
- Aidan LaBella. **Optimized Flight Safety Event Detection in the NGAFID**. *The 2022 RIT Undergraduate Research Symposium*. Rochester, NY. July 28, 2022.
- Aidan LaBella. **Data Mining in the National General Aviation Flight Information Database**. *DataFest 2023*. Rochester, NY. March 24, 2023.

SERVICE

- **Computer Science Department Representative** Brown University
Graduate Student Council *Sept 2023 - Present*
 - Participate in monthly meetings to discuss and vote on issues pertaining to graduate students at Brown.

MENTORING

- **SWIFT-SAT: RFI Detection Across Six Orders of Magnitude in Intensity** Brown University
Scientific Machine Learning Sept 2023 - Present
 - **Kevin Scroggins, Jr.:** Sc.M., Computer Science
 - **Christopher Tripp:** Sc.M., Computer Science
 - **Ronit Kapoor:** Sc.M., Computer Science
 - **Sarah Liu:** Sc.B., Computer Science
 - **Charles Duong:** Sc.B., Computer Science
- **GATS: Time Series Data Analysis for General Aviation with Self-Supervised Learning** Brown University
Machine Learning for General Aviation Safety Sept 2024 - Present
 - **Elise Carman:** Sc.M., Computer Science
 - **Justin Long:** Sc.M., Computer Science
 - **Nathan DePiero:** Sc.M., Computer Science
 - **Charles Duong:** Sc.B., Computer Science